OUTPUT SCREENSHOTS:

```
Number of student records: 1000
Linear Search: Index = 339 Comparisons = 340 Time = 0.0599 ms
Iterative Binary Search: Index = 500 Comparisons = 9 Time = 0.0157 ms
Recursive Binary Search: Index = 500 Comparisons = 9 Time = 0.0209 ms

Number of student records: 10000
Linear Search: Index = 8802 Comparisons = 8803 Time = 0.8594 ms
Iterative Binary Search: Index = 5000 Comparisons = 13 Time = 0.0164 ms
Recursive Binary Search: Index = 5000 Comparisons = 13 Time = 0.0279 ms

Number of student records: 100000
Linear Search: Index = 86324 Comparisons = 86325 Time = 14.3668 ms
Iterative Binary Search: Index = 50000 Comparisons = 16 Time = 0.0181 ms
Recursive Binary Search: Index = 50000 Comparisons = 16 Time = 0.0288 ms

Number of student records: 1000000
Linear Search: Index = 814170 Comparisons = 814171 Time = 190.5971 ms
Iterative Binary Search: Index = 500000 Comparisons = 19 Time = 0.0179 ms
Recursive Binary Search: Index = 500000 Comparisons = 19 Time = 0.0265 ms
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/Arun/Documents/daa.py ==========

Dataset size: 1000
Linear Search | Average time: 0.038403 ms | Average comparisons: 479.3
Iterative Binary Search | Average time: 0.002773 ms | Average comparisons: 9
Recursive Binary Search | Average time: 0.003797 ms | Average comparisons: 9

Dataset size: 10000
Linear Search | Average time: 0.432937 ms | Average comparisons: 4896.93
Iterative Binary Search | Average time: 0.003967 ms | Average comparisons: 12.2
Recursive Binary Search | Average time: 0.005137 ms | Average comparisons: 12.2

Dataset size: 100000
Linear Search | Average time: 8.397403 ms | Average comparisons: 51220.87
Iterative Binary Search | Average time: 0.005983 ms | Average comparisons: 15.33
Recursive Binary Search | Average time: 0.00748 ms | Average comparisons: 15.33

Dataset size: 1000000
Linear Search | Average time: 131.864017 ms | Average comparisons: 521474.03
Iterative Binary Search | Average time: 0.009707 ms | Average comparisons: 18.93
Recursive Binary Search | Average time: 0.013133 ms | Average comparisons: 18.93
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================== RESTART: C:/Users/Arun/Documents/Daa 123.py ==================

Number of student records: 1000
Linear Search: Index = 6 Comparisons = 7 Time = 0.0172 ms
Iterative Binary Search: Index = 500 Comparisons = 9 Time = 0.0171 ms
Recursive Binary Search: Index = 500 Comparisons = 9 Time = 0.0266 ms

Number of student records: 10000
Linear Search: Index = 491 Comparisons = 492 Time = 0.0513 ms
Iterative Binary Search: Index = 5000 Comparisons = 13 Time = 0.0152 ms
Recursive Binary Search: Index = 5000 Comparisons = 13 Time = 0.0188 ms

Number of student records: 100000
Linear Search: Index = 98668 Comparisons = 98669 Time = 17.5234 ms
Iterative Binary Search: Index = 50000 Comparisons = 16 Time = 0.0137 ms
Recursive Binary Search: Index = 50000 Comparisons = 16 Time = 0.0172 ms

Number of student records: 1000000
Linear Search: Index = 441847 Comparisons = 441848 Time = 105.9797 ms
Iterative Binary Search: Index = 500000 Comparisons = 19 Time = 0.0156 ms
Recursive Binary Search: Index = 500000 Comparisons = 19 Time = 0.0244 ms
```